

Exploring the Relationship Between Player Stats, Team Performance, and Game Results in the NBA

Project Subject Area

This project explores the relationship between individual NBA player performance, team standings, and fan engagement metrics. The goal is to analyze how on-court performance and public visibility contribute to a team's overall success in the league using a combination of datasets from a flat file, website, and API.

Data Sources

Flat File – Player Performance Data

Description: This dataset contains detailed player statistics for NBA players, including points per game, assists, rebounds, and shooting percentages.

Source: Kaggle – NBA Player Stats Dataset

Link: <https://www.kaggle.com/datasets/drgilermo/nba-players-stats>

File Type: CSV

Website – Team Performance & Rankings

Description: This dataset provides official NBA standings including win-loss records, win percentages, and conference rankings.

Source: Basketball Reference – NBA Standings

Link: https://www.basketball-reference.com/leagues/NBA_2023_standings.html

API – Game and Player Data

Description: This API provides access to player details, team information, and game-level stats like scores, matchups, and dates.

Source: balldontlie.io

Link: <https://www.balldontlie.io/>

Relationships

All datasets are connected through the Team Name or Team ID field:

- The Player Stats dataset is linked to team-level data by Team Name.
- The Team Standings dataset directly tracks each team's win-loss record and rank.
- The API provides team and player metadata and game results that map to both sources.

Combining these will allow me to build a unified dataset analyzing how individual performance and team performance correlate with actual game outcomes and success metrics.

Project Approach/Plan

This project is designed to explore the relationship between NBA player performance, team standings, and fan engagement. By combining structured player statistics from a CSV file, official team standings from a scraped website, and game results and player metadata from the balldontlie.io API, my goal is to produce a well-rounded view of team performance and success.

My approach will begin with data collection and cleaning. Player performance stats from the Kaggle CSV file will be grouped by team to calculate average points, assists, rebounds, and ball shooting. The Basketball Reference website will be scraped for wins, losses, and team rankings. The API will be used to get game results and player game level contributions, which will improve the dataset with current or historical performance.

All datasets will be joined using the Team Name or a Team ID. After merging, I will do an exploratory data analysis to identify correlations and patterns. I may look into whether higher fan-followed teams tend to perform better, or whether teams with more consistent player stats rank higher in standings.

Tools like Pandas, Matplotlib, and Seaborn will support my analysis.

Concerns and Challenges

I anticipate a few challenges throughout this project. My first concern is data inconsistency with team naming across the CSV, website, and API. I'll fix this with mapping dictionaries and standardization functions. Second, the balldontlie API has rate limits, so data will need to be requested in manageable batches or cached locally. A third concern is handling mid-season trades. Players might contribute to multiple teams in a season, complicating performances grouped by teams. I will account for this by filtering game stats by date ranges and team affiliation during that time. Also, scraping from Basketball Reference may be sensitive to layout changes, which would need adjustments.

Ethical Implications

This project uses publicly available, non-personal data. No private or sensitive information is collected or exposed. It's still important to be aware of potential biases in interpreting correlations. For example, fan engagement may not directly cause better performance, even if a correlation exists. I must be careful not to make unsupported causal claims. I will also respect the usage policies of all data sources. All results will be presented objectively and transparently, and all data transformations will be documented.